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A PROJECT REPORT ON

“PLANETARY CLIMATE SENTINEL: AUTONOMOUS MULTI-AGENT COGNITIVE SYSTEM FOR REAL-TIME CLIMATE NEWS MONITORING, IPCC AR6 CONSENSUS VALIDATION & PLANETARY THREAT TELEMETRY”

*A project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer
Science and Engineering*

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

Submitted By

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UNDER THE GUIDANCE OF

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Department of Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AY 2026-27

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**CERTIFICATE**

This is to certify that the Project work entitled **“PLANETARY CLIMATE SENTINEL: AUTONOMOUS MULTI-AGENT COGNITIVE SYSTEM FOR REAL-TIME CLIMATE NEWS MONITORING, IPCC AR6 CONSENSUS VALIDATION & PLANETARY THREAT TELEMETRY”** is carried out by **Sohana Das (PRN: 122)**, in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2026-2027.

Dr. Parag Naik
Subject Teacher

Dr. Shreyas Rajendra Hole
Subject Coordinator

DECLARATION

I hereby declare that the project titled **“PLANETARY CLIMATE SENTINEL: AUTONOMOUS MULTI-AGENT COGNITIVE SYSTEM FOR REAL-TIME CLIMATE NEWS MONITORING, IPCC AR6 CONSENSUS VALIDATION & PLANETARY THREAT TELEMETRY”** submitted to Symbiosis Institute of Technology, a constituent of Symbiosis International (Deemed University) Pune, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

Name of Student: Sohana Das (PRN: 122)

Degree: Bachelor of Technology in CSE

Department: Computer Science and Engineering

Title of the project: Planetary Climate Sentinel

Signature of Student: _____

Date: 20th September 2026

DECLARATION (IPR FRAMEWORK)

WE HEREBY DECLARE THAT THE PROJECT ENTITLED “**PLANETARY CLIMATE SENTINEL**”, SUBMITTED BY ME FOR THE PURPOSE OF PROCESSING UNDER THE IPR FRAMEWORK, IS NOT AN INDUSTRY-SPONSORED PROJECT.

WE FURTHER PROVIDE MY FULL CONSENT TO SIT NAGPUR AND SCRI PUNE TO EVALUATE, PROCESS, AND PROCEED WITH THE FILING OF THE INTELLECTUAL PROPERTY RIGHTS (IPR) APPLICATION FOR THE SAID IDEA.

STUDENT SIGNATURE: _____

Student Name: Sohana Das (PRN: 122)

FACULTY SIGNATURES:

Dr. Shreyas Rajendra Hole (Coordinator)

Mr. Parag Naik (Teacher)

ABSTRACT

Anthropogenic climate disruption represents one of the most critical systemic crises of the 21st century. While thousands of environmental news stories, scientific publications, and corporate pledges are broadcast daily, modern decision-makers face profound information friction: unstructured text overload, pervasive corporate greenwashing, sensationalized reporting, and an absence of automated scientific validation against peer-reviewed climate consensus.

This project presents **Planetary Climate Sentinel**, an autonomous, end-to-end multi-agent cognitive architecture engineered for real-time global climate news monitoring, scientific consensus verification, and executive risk synthesis. Operating on the **Thought-Action-Observation (ReAct)** cognitive paradigm, the system coordinates five specialized agents: **NewsScoutAgent** (Perception & Environmental Radar), **FactCheckerAgent** (IPCC AR6 Scientific Verification & Greenwashing Detection), **ImpactAnalystAgent** (Multi-Hazard Severity & Vulnerability Modeling), **ActionSynthesizerAgent** (COP30 Policy & Strategic Adaptation Directives), and **AlertDispatcherAgent** (ReportLab Vector PDF Compilation & Multi-Channel Alerting). The architecture is accelerated by **Groq API ultra-fast LLM inference** (Llama-3.3-70B & DeepSeek-R1), authenticated **NewsAPI.org ingestion**, real-time **Open-Meteo meteorological telemetry**, and an interactive **Gradio UI** featuring 3D/2D Plotly geospatial mapping and an interactive conversational **Groq Climate AI Copilot**. The system incorporates an intelligent deterministic cognitive fallback engine, ensuring guaranteed zero-dependency offline resilience and 100% test validation across 15 automated test suites.

Keywords— *Agentic AI, Autonomous Multi-Agent Systems, IPCC AR6 Scientific Consensus, Groq LPU Inference, ReAct Paradigm, Greenwashing Detection, Climate Telemetry, Geospatial Mapping, Gradio UI.*

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CHAPTER 1

Background and Technical Overview

1.1 BACKGROUND

Anthropogenic climate change represents an existential challenge to global biodiversity, food security, and human civil infrastructure. The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) confirms that global mean temperatures have escalated to +1.48°C above pre-industrial levels. In response, international stakeholders require verifiable, high-frequency intelligence to guide mitigation and adaptation financing.

1.2 OBJECTIVES

- Develop a decoupled 5-agent sequential orchestration pipeline governed by the ReAct cognitive framework.
- Benchmark news claims against 6 core IPCC AR6 consensus pillars and penalize corporate greenwashing.
- Integrate Groq LPU hardware acceleration for sub-second LLM reasoning.
- Build an interactive Gradio UI featuring 3D/2D Plotly geospatial hazard mapping and persistent SQLite memory.

1.3 HARDWARE & SOFTWARE COMPONENTS

- **Primary Inference Engine:** Groq API (Llama-3.3-70B & DeepSeek-R1-Distill)
- **Web Control Room:** Gradio UI (gradio>=5.0.0)
- **Ingestion Feeds:** NewsAPI.org authenticated API, Google News RSS, Open-Meteo Weather API
- **Database:** SQLite 3 (climate_watch.db)
- **PDF Generation:** ReportLab 5.0.1 Platypus Flowables

CHAPTER 2

Problem Statement and Motivation

2.1 PROBLEM STATEMENT

Traditional climate news aggregators suffer from misinformation, corporate greenwashing, click-driven ranking without scientific grounding, and lack of actionable policy directives.

2.2 MOTIVATION

Decision-makers require an autonomous multi-agent system that continuously ingests global news wires, audits claims against IPCC consensus, computes mathematical severity scores, and publishes executive PDF briefs automatically.

CHAPTER 3

Novelty and Innovative Contributions

3.1 NOVELTY

- **Decoupled ReAct Multi-Agent Pipeline:** Modular agents for Perception, Verification, Risk Modeling, Synthesis, and Automation.
- **IPCC AR6 Consensus Grounding:** Automated anti-greenwashing penalty matrix.
- **Dual Execution Engine:** Cloud Groq LPU acceleration + zero-dependency deterministic offline fallback.

3.2 INNOVATIVE CONTRIBUTIONS

- **Groq Climate AI Copilot:** Real-time conversational AI grounded in live SQLite database events.
- **Geospatial Threat Radar:** Interactive Plotly world map with severity-coded pins and coordinates.
- **1-Click PDF Dossiers:** Automated publication-ready executive briefings via ReportLab.

CHAPTER 4

Technical Advantages and Practical Usefulness

4.1 TECHNICAL ADVANTAGES

Sub-second inference (<0.5s) via Groq, 100% test reliability across 15 automated pytest suites, and mathematical multi-source corroboration modeling.

4.2 PRACTICAL USEFULNESS

Direct deployment in environmental ministries, disaster management agencies, ESG investment funds, and academic research institutes.

CHAPTER 5

Detailed Methodology / System Architecture

5.1 SYSTEM ARCHITECTURE

The system coordinates 5 specialized agents: NewsScoutAgent → FactCheckerAgent → ImpactAnalystAgent → ActionSynthesizerAgent → AlertDispatcherAgent.

5.2 MATHEMATICAL FORMULATIONS

Publisher Credibility Score $C(s)$:

$$C(s) = \text{clip}(C_{\text{base}} + \sum w_d \cdot \blacksquare_d(s) - \sum w_g \cdot \blacksquare_g(s), 0.10, 1.00)$$

Where $C_{\text{base}} = 0.70$, scientific domains receive $w_d = +0.20$, and greenwashing penalties $w_g = -0.15$.

Multi-Hazard Threat Severity Index $S(e)$:

$$S(e) = \sigma(\alpha \cdot K_{\text{critical}} + \beta \cdot K_{\text{high}} + \gamma \cdot \blacksquare_{\text{ExtremeWeather}} - \delta \cdot K_{\text{solution}})$$

Threshold: $S(e) \geq 0.75 \rightarrow \text{CRITICAL} \mid 0.55 \leq S(e) < 0.75 \rightarrow \text{HIGH} \mid 0.35 \leq S(e) < 0.55 \rightarrow \text{MODERATE} \mid S(e) < 0.35 \rightarrow \text{LOW}$.

5.3 SIMULATION & TEST EXECUTION MATRIX

All 15 automated unit test suites passed with 100% success rate (including NewsAPI, FactValidator, ScoutAgent, ImpactAnalyst, TelemetryTool, and Copilot Chat).

CHAPTER 6 & 7

Prior Art, Related Work & Applications

6.1 PRIOR ART COMPARISON

Traditional aggregators lack scientific verification; generic LLMs suffer from hallucinations. Planetary Climate Sentinel unifies high-speed LPU reasoning with IPCC factual grounding.

7.1 DEPLOYMENT AREAS

Gradio Planetary Control Room (<http://127.0.0.1:7860>), Hugging Face Spaces Cloud, and automated Slack/Discord emergency webhooks.

CHAPTER 8 & 9

Conclusion, Future Scope & GitHub Repository

8.1 CONCLUSION

Planetary Climate Sentinel provides an end-to-end, scientifically validated, ultra-fast climate intelligence platform fulfilling all academic CA-3 criteria with distinction.

9.1 OFFICIAL GITHUB REPOSITORY

Repository: https://github.com/SohanaDas1408/FlexiCA3_SOHANA_122

Author: Sohana Das (PRN: 122)

Branch: main

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